Www.Medicalstudyzone.com

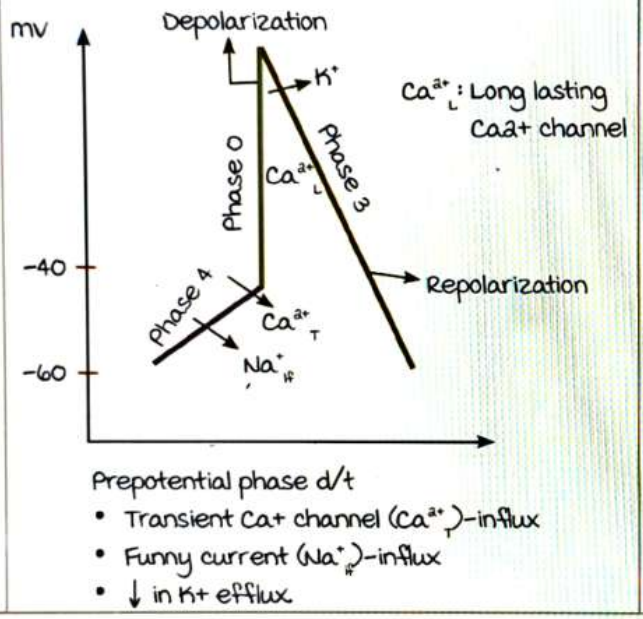
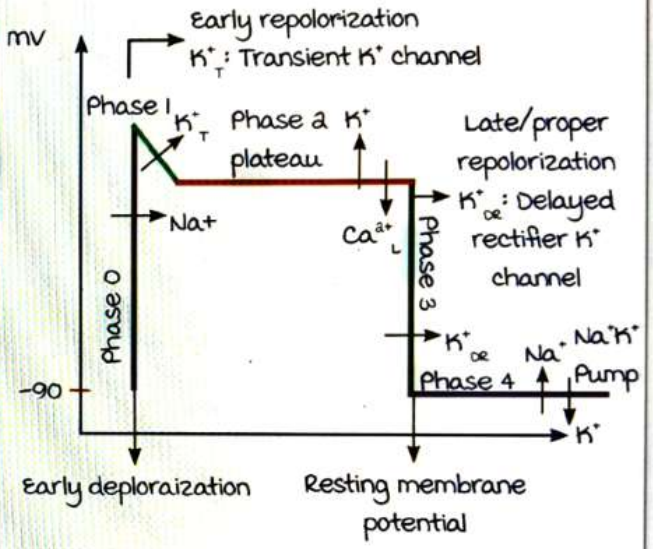
----- Active space -----

CARDIAC ACTION POTENTIALS

SA Nodal & Ventricular Action Potential

00:02:09

Comparison b/w SA nodal & Ventricular Action Potential:

	SA Nodal Potential (Pacemaker)	Ventricular Action Potential
RMP	Restless membrane potential = -60 to -40 mV	Resting membrane potential = -90 mV
Property	Chronotropy (Heart rate generation)	Inotropy (Contractility)
Action potential		

Funny currents/hyperpolarized cyclic nucleotide channels (HCN):

Location : Eye, Heart (SA node).

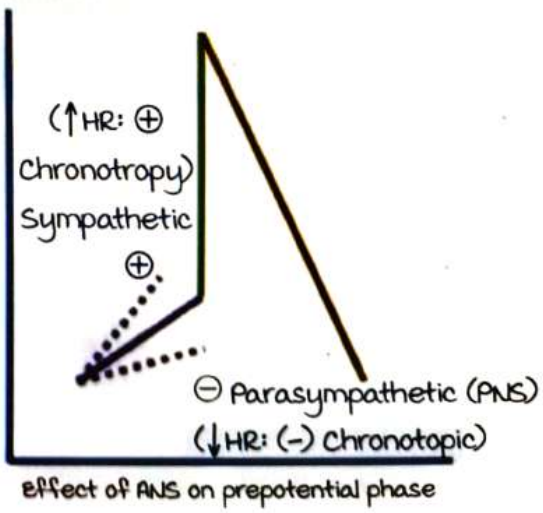
Opened by : Cyclic GMP.

Function : Vision, heart rate generation.

Applied aspect :

Ivabradine : Funny current blocker.

- Used in Rx of arrhythmia.
- S/E : ↓ Visual acuity.



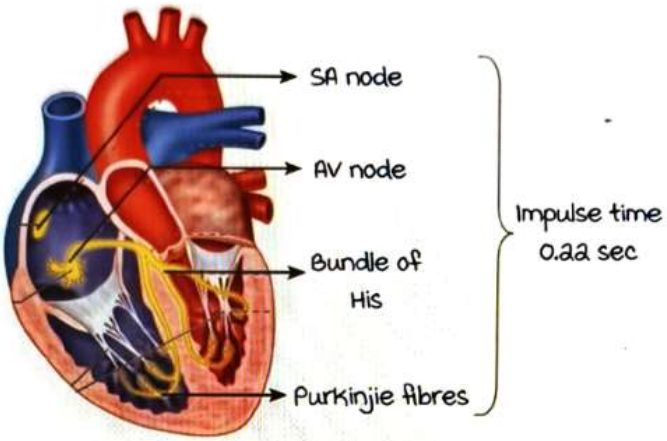
Effect of ANS on prepotential phase

Www.Medicalstudyzone.com

Conduction System of Heart

00:26:39 ----- Active space -----

Dromotropy : Conduction of impulses.



1. SA node :

- Dominant pacemaker.
- Impulse : 100/min (Highest rate).
  - Resting vagal tone $\xrightarrow{\ominus}$ SA node → HR : 72/min.
  - Atropine (PNS blocker) $\xrightarrow{\oplus}$ SA node → HR : 150 bpm.
  - Heart transplant (Denervated heart) $\xrightarrow{\text{No effect}}$ SA node → HR : 100 bpm.
- Overdrive Suppression : Suppresses impulses generated by other pacemakers in the heart.

2. AV node :

- Gatekeeper of heart.
- Physiological delay of impulse conduction : 0.1 sec.
  - For adequate ventricular relaxation.
- Conduction velocity : 0.04 m/s – slowest (D/t ↓ gap junctions).

3. Bundle of His.

4. Purkinje fibres :

- Innervates every single ventricular myocardial cell.
  - Functional syncytium : Synchronised contractions.
- Conduction velocity : 4 m/s – highest (Due to ↑ gap junctions at intercalated disc).

Feedback